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MIL-L-24435(SHIPS)
8 September 1970

MILITARY SPECIFICATION

LUBRICATING OIL, MARINE TWO STROKE CYCLE GASOLINE ENGINE

1. SCOPE

1.1 This specification covers one grade of lubricating oil (Military Symbol OB-40), suitable for two-cycle water cooled gasoline engines using fuel conforming to MIL-G-3056. This lubricating oil is intended to be mixed with the fuel at the ratio prescribed by the engine manufacturer prior to introduction into the engine.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
RR-S-366 - Sieve, Test.
NNN-C-940 - Cylinder, Graduated, Laboratory, Glass.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-290 - Packaging, Packing, and Marking of Petroleum and Related Products.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 92 - Method of Test for Flash and Fire-Points by Cleveland Open Cup.
D 97 - Test for Pour Points.
D 129 - Test for Sulfur in Petroleum Products by the Bomb Method.
D 270 - Petroleum and Petroleum Product, Sampling.
D 287 - Method of Test for API Gravity of Crude Petroleum, Petroleum and Petroleum Products (Hydrometer Method).
D 445 - Method of Test for Viscosity of Transparent and Opaque Liquids (Kinematic and Dynamic Viscosities).
D 811 - Chemical Analysis of Metals in New and Used Lubricating Oils.
D 874 - Method of Test for Sulfated Ash from Lubricating Oils and Additives.
D 1091 - Method of Test for Phosphorus in Lubricating Oils and Additives.
D 1748-62T - Rust Protection by Metal Preservatives in the Humidity Cabinet.
D 2270 - Calculating Viscosity Index from Kinematic Viscosity.

(Application for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. The lubricating oil furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

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3.2 Material. The oil shall be essentially a stable homogeneous blend of petroleum-base lubricating oil stocks plus additive agents, as necessary, to conform to this specification.

3.3 Lubrication. The oil shall satisfactorily lubricate oil-lubricated two-cycle gasoline engine parts when mixed with fuel at the engine manufacturer's prescribed fuel to oil ratio.

3.4 Performance. The oil shall not cause either excessive ring sticking or clogging of oil channels, shall keep cylinder and ring wear down to a minimum and shall not cause excessive lacquer, carbon or sludge deposits on any part of the engine in which it is intended for use. Satisfactory characteristics of the oil in this respect shall be demonstrated when the oil is tested as specified in the Appendix.

3.4.1 When the oil is tested as specified in the appendix the rating assigned the engine parts shall not exceed the following:

Characteristic	Limit
Piston varnish (10 = clean)	8.5 1/
Piston scuffing	None
Piston crown deposit (max.)	100 percent medium 1/
Piston ring sticking (10 = free)	9.5 1/
Piston ring groove deposits	Report
Cylinderhead deposit (max.)	100 percent medium 1/
Exhaust port plugging (max.)	5 percent 1/
Exhaust passage plugging (max.)	5 percent
Preignition	None/no damage
Sparkplugs fouled (max.)	2/cylinder
Piston ring condition	No scuffing/heavy scoring
Top piston ring wear (max.)	0.02 grams 1/
Cylinder wear (across exhaust bridges)	0.002 inches 1/
Bearing condition	Good (light blue discoloration permitted)
Cylinder condition	Good - No scuffing/scoring

3.5 The oil shall conform to the chemical and physical requirements shown in table I.

Table I - Chemical and physical requirements.

Characteristics	Requirements	Test Method
Gravity, Degree A.P.I.	Report	D 287
Sulfur	Report	D 129
Viscosity at 210°F. Centistokes	11.9 - 15.5	D 445
Viscosity index	Report	D 2270
Flash point, °F., min.	390	D 92
Pour point, °F., max.	10	D 97
Ash, sulfated, percent, min,	Report	D 874
Rust protection	pass	(see 4.6.2)
Miscibility	pass	(see 4.6.3)
Water tolerance	pass	(see 4.6.4)
Homogeneity	pass	(see 4.6.5)
Contamination (mg./gal. max.)	10	(see 4.6.6)
Contamination (fibers/gal. max.)	1	(see 4.6.6)
Metal content	1/	D 811
Phosphorus	1/	D 1091
Nitrogen	1/	Kjeldahl

1/ Identify quantitatively for record purposes.

1/ Average of all similar parts in the engine.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the tests specified in 4.6. and the appendix. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3, 6.3.1, 6.3.2, and 6.3.3).

4.2.1 Four copies of the test report together with one copy of the engine test log and the engine parts listed in 70.2.1.13 of the appendix, except the cylinder block and crankshaft shall be forwarded for evaluation to the Naval Ship Engineering Center, Center Building, Prince George's Center, Hyattsville, Maryland 20782. The cylinder block shall be retained at the testing facility until qualification action has been taken.

4.2.2 One 4-ounce sample of the base oil blend and one 4-ounce sample of the finished oil which was engine tested in accordance with the appendix are required for infrared analysis which is for record purposes. The two samples shall be forwarded to the Naval Ship Engineering Center, Philadelphia Division, Philadelphia, Pennsylvania 19112.

4.3 Sampling for quality conformance.

4.3.1 Lot. For purposes of sampling, a lot shall consist of all lubricating oil manufactured as one batch, and offered for delivery at one time.

4.3.2 Sampling for examination of filled containers. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105 at inspection level I and acceptable quality level (AQL) = 2.5 percent defective to verify compliance with all stipulations of this specification regarding fill, closure, marking; and other requirements not involving tests.

4.3.3 Sampling for tests.

4.3.3.1 Blending tank. A composite sample of not less than 5 gallons of oil shall be taken from each lot in accordance with the applicable procedures of ASTM D 270.

4.4 Examination of filled containers. Each sample filled container selected in accordance with 4.3.2 shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample, having one or more defects, or under required fill, shall be rejected and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.5 Quality conformance tests. The sample specimen selected in accordance with 4.3.3.1 shall be subjected to the tests specified in 4.6 except the test of 4.6.5. To insure that the material is essentially the same as that given qualification under this specification, in addition to conforming with this specification, the results of quality conformance tests listed below shall not vary from the analysis of the sample given qualification to an extent greater than the tolerances shown. The analysis referred to is that which was obtained at the laboratory upon the sample that was evaluated for qualification under this specification.

Gravity, degrees A.P.I.	+1.0 degree.
Ash, sulfated, percent	No minus.
Flash point °F.	+25°F.
Sulfur, percent	∓20 percent.
Viscosity, Saybolt Seconds Universal (SSU) at 210°F.	∓3 percent.
Pour point °F.	∓5°

4.5.1 Rejection of lots. When the sample of lubricating oil fails any of the tests, the entire lot shall be rejected.

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4.6 Test procedures.

4.6.1 Properties. The tests and methods used to determine the physical and chemical properties of the oil shall be as specified in table I.

4.6.2 Rust protection. Steel test panels shall be prepared as specified in Section 7, 8a, and 8b of ASTM D 1748-62T except that an oil temperature of 75±5°F. shall be used in lieu of the temperature specified in Section 8a and a drain time of 10 minutes shall be used in lieu of the drain time specified in Section 8b.

4.6.2.1 After the panels have been dipped and drained they shall be immersed vertically in a 75±5°F. solution of 1/2 pound chemically pure sodium chloride per gallon of distilled water. The panels shall show no visible rust except for the area within 1/8 inch of the panel edges or within 1/8 inch of any holes after 8 hours immersion. A color photograph of the panel after completion of the test shall be included in the qualification test report.

4.6.3 Lubricant miscibility with fuel. A 500 ml. graduated cylinder conforming to NNN-C-940 shall be used for the miscibility determination. The cylinder, oil, and gasoline to be used shall be brought to a temperature of 32°F. and maintained at that temperature throughout the test. First add 17 ml. of the oil to be tested to the cylinder and then add sufficient gasoline to bring the total volume of fluid in the cylinder to 450 ml. Mix the oil and gasoline by alternately inverting and returning the closed container to an upright position. Record the number of inversions necessary to completely mix the oil and gasoline. The mixture is considered complete if no oil remains clinging to the walls of the cylinder. A pass rating is given if the oil and gasoline are completely mixed in not more than 24 inversions within two minutes elapsed time.

4.6.4 Water tolerance. The water tolerance shall be determined by each of the following procedures:

(a) Procedure A - Add 75.0 ml. oil to a 100 ml. centrifuge tube. Add 1.0 ml. of distilled water. Shake 2 minutes at room temperature. Allow to stand 20 hours at room temperature. Evaluate.
(b) Procedure B - Add 75.0 ml. oil to a 100 ml. centrifuge tube. Add 0.15 ml. of distilled water. Shake 2 minutes at room temperature. Evaluate.
(c) Procedure C - Same as Procedure B, except that after 20 hours standing, the tube is centrifuged 40 minutes @ 600-700 revolutions per minute (r.p.m.) and then evaluated.

The water tolerance of the oil is considered satisfactory if no emulsion or sediment is present at evaluation. A color photograph of the three tubes shall be included in the qualification test report.

4.6.5 Homogeneity.

4.6.5.1 Approximately 100 ml. of the test oil, in a pour-point test jar, shall be cooled to minus 25°F. and held at this temperature for 24 hours, then allowed to warm to room temperature. The test oil shall then be examined for separation of components.

4.6.5.2 The test oil, when tested in accordance with 4.6.5.2.1 shall not exhibit sedimentation or separation of insoluble materials to an extent greater than 0.01 percent by volume.

4.6.5.2.1 Procedure. Two calibrated 100 ml. centrifuge tubes shall be filled with 100 ml. each of the test lubricating oil. These two samples shall then be centrifuged for four hours in a DeLaval 100 ml. test tube centrifuge, at 6000 r.p.m. and 100°±2°F. Other centrifuges may be used, provided the product of the relative centrifugal force at the tube tips and the time of centrifuging equals 2,400,000.

4.6.6 Contamination. A four-gallon representative sample shall be used for this test. A quantity of clear, unleaded gasoline or naphtha shall be filtered through the sieve to insure cleanliness. A 200-mesh sieve conforming to RR-S-366 shall be used, except that the sieve frame shall have a nominal 3 inch diameter. The sieve shall be dried at 100°C. (212°F.) and cooled in a desiccator and shall then be weighed to the nearest milligram. The entire oil sample (4-gallons) together with an equal amount of filtered naphtha or gasoline shall be mixed thoroughly at room temperature and filtered through the weighed sieve. After filtration of the oil, the oil container shall be thoroughly rinsed with clean naphtha or gasoline and the washings filtered through the same weighed sieve. The sieve shall then be thoroughly washed with filtered naphtha or gasoline and dried at 100°C. (212°F.) to remove

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the naphtha, or gasoline. The sieve shall again be cooled in a desiccator and weighed to the nearest milligram. The increase in weight of the sieve shall not exceed 40 milligrams (0.040 gram) and the presence of fibrous material in the residue shall be reported. A fiber shall be defined as a thread like structure with a maximum dimension exceeding six millimeters.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Packaging and packing. Two-cycle engine lubricating oil shall be packaged and packed the level specified (see 6.2) in accordance with MIL-STD-290. The type and size containers shall be as specified by the procuring activity (see 6.2). All materials used in the construction of the containers shall be such as will not affect or be affected by the contained two-cycle engine lubricating oil. Before filling, all containers shall be thoroughly cleaned and inspected to insure absolute absence of dirt, corrosion products, water, or other materials which would contaminate or interfere with the satisfactory performance of the oil.

5.2 Marking of shipments. Shipping containers shall be marked in accordance with MIL-STD-290. The nomenclature shall be as follows:

Lubricating Oil, Marine Two-Cycle Gasoline Engine, Military Symbol OB-40, Specification MIL-L-24435(SHIPS), Manufacturer's Name, Batch Number, Qualification Test Reference Number, and Contractor's Order Number.

6. NOTES

6.1 Intended use. The lubricating oil covered by this specification is intended for use in various models of watercooled two-cycle gasoline engines (using fuel conforming to MIL-G-3056) when mixed with the fuel at the engine manufacturer's prescribed ratio.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) The type and size of container required (see 5.1).
(c) Selection of the applicable level of packaging and packing required (see 5.1).
(d) Quantity in gallons. The material should be purchased by volume, the unit being a U.S. gallon at 15.6°C. (60°F.).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-24435 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Center Building, Prince George's Center, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.3.2 Requests for test of oil will be considered only under the conditions that the manufacturer submit the following information regarding composition of the oil proposed for qualification:

(a) Crude source, general manufacturing process for each stock and type of base or blending stocks and percentages used in the finished oil.
(b) Type of additive agents, typical chemical analysis and percentages used in the finished oil.
(c) Typical analysis of the blending stocks, the base oil and the finished oil, including percentage of metallic elements contributed by the additive.

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6.3.3 Each sample of oil, rebrand or experimental, submitted for qualification shall be designated by the exhibitor by a different name, number, or letter to denote any change in base oil components or the amounts or type of additive percent.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 9150-N321)

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APPENDIX

PERFORMANCE CHARACTERISTICS OF TWO-CYCLE

GASOLINE ENGINE LUBRICATING OILS

10. SCOPE

10.1 This method describes a test procedure for evaluating the performance characteristics of two-cycle oils under severe operating conditions using a relatively poor quality fuel.

10.2 Summary. A water cooled two-cycle gasoline engine with a manufacturers rating of not less than 40 brake horsepower is operated with a test oil for 147 hours under cyclic conditions. The performance of the test oil is indicated by the condition of the engine parts including piston ring sticking, engine deposits, and engine wear.

20. APPLICABLE DOCUMENTS

20.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
P-D-680 - Dry Cleaning Solvent.

MILITARY
MIL-G-3056 - Gasoline, Automotive, Combat.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. SAMPLE

30.1 The quantity of test oil required varies with the engine used, but generally ranges from 14-25 gallons.

40. APPARATUS

40.1 Engine. The engine used in the test shall be a water-cooled, two-cycle gasoline engine with a manufacturers rating of at least 40 brake horsepower. The engine shall not be fitted with a capacitor discharge ignition system and surface gap spark plugs, but, shall be fitted with suitable adjustable carburetor jets.

40.1.1 Engine loading. The engine shall be loaded so that the engine speed is maintained near the mid point of the full power operating range recommended by the manufacturer. The speed variation shall be no more than ±200 rpm. The more satisfactory means of loading is by use of a suitable test wheel; however, a dynamometer may be used if all the test conditions specified herein are met.

40.2 Test tank. There are no restrictions on the design of the test tank provided the tank is suitable for maintaining the test conditions specified herein and provided the tank's exhaust system prevents the exhaust from entering the carburetor.

40.3 Instrumentation.

40.3.1 Fuel flow measurement. Any design of fuel flow measuring instrumentation is acceptable provided it is suitably calibrated.

40.3.2 Temperature measurement. Thermocouples shall be used to measure the following variables:

Variable	Location of thermocouple
Cooling water-out temperature	Discharge passage of thermostat housing
Spark plug gasket temperatures 2/	Base of each spark plug
Test tank water temperature	Lower front corner of test tank

2/ The spark plug gasket temperatures shall be continuously monitored by a recording potentiometer throughout the test.

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40.3.3 Engine speed. A tachometer with an appropriate range shall be used to measure engine rpm. An automatic engine overspeed control may be used.

50 MATERIALS

50.1 Fuel. Only fuels accepted by the Naval Ship Engineering Center, Center Building, Prince George's Center, Hyattsville, Maryland 20782 may be used for final acceptance of an oil. To be accepted the fuel must be in accordance with type I of MIL-G-3056 and show a satisfactory severity[3/] in an engine performance test. Acceptance will apply to all batches of fuel made by a refiner from the same base stocks providing they remain within the tolerances established for the chemical and physical characteristics.

50.2 Stoddard Solvent in accordance with P-D-680.

50.3 Solvent for cylinder block cleaning[4/].

50.4 Paper deglazer[5/].

60 PROCEDURE

60.1 Engine reconditioning and assembly. A complete new power head assembly should be used for each test. However, if the cylinder block is to be re-used the following new parts shall be used in the power head build-up:

Part Description
Cylinder heads(s)
Piston
Piston ring sets
Wrist pins
Wrist pin bearing assembly(s)
Connecting rod bearings
Connecting rod bearing retainers
Top main bearing assembly
Center main bearing and cage
Center main bearing race
Bottom main bearing
All crankshaft seals
O Rings
Lower crankcase gasket
Power Head gasket set (including head gaskets)
Breaker points
Condenser
Spark plugs

60.1.1 Cleaning procedure. The re-used parts shall be cleaned as follows:

60.1.1.1 Cylinder block. Scrape and wire brush the cylinder block free of carbon, paying particular attention to the exhaust parts and passages. Inspect cylinder walls for scoring and then deglaze with deglazer paper. Soak cylinder block in cleaning solvent (see 50.3) until clean and then rinse for 5 minutes with hot water.

60.1.1.2 Other parts. Inspect crankshaft and connecting rods for rust pits and score marks and then spray wash with Stoddard Solvent, (see 50.2). Clean all other parts with Stoddard Solvent.

60.1.1.3 Only the oil to be tested or a qualified product under this specification may be used for lubrication during assembly of a new or rebuilt power head.

3/ A fuel that has been found satisfactory is American Oil Company's Certified Fuel MS-08.

4/ A material found satisfactory for this purpose is Cities Service S-26, which is manufactured by Cities Service Oil Company.

5/ A material found satisfactory is 3-M emery code 300A wet or dry which is maunfactured by 3-M Company.

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60.1.2 Engine measurements. The following measurements shall be taken and recorded during the power head assembly:

Measurement
Cylinder bore 1/4 inch below top ring travel at 0° and 90° crankshaft
Cylinder bore across exhaust bridges
Cylinder bore across intake bridges
Piston to cylinder fit at 0° and 90° crankshaft
Piston to wrist pin fit at 0° and 90°
Piston ring side clearance
Piston ring butt gap (in cylinder)
Piston ring weight in grams

New power head assemblies shall be disassembled and the above measurements taken.

60.1.3 Power head tolerances. Power head assemblies (both new and re-built) to be used for testing shall meet all manufacturer's tolerances.

60.1.4 Compression ratio. Calculate the compression ratio from the following relationship:

$$\text{Compression ratio} = \frac{D+C}{C}$$

Where:
D = displacement or effective swept volume of the piston (determined for engine measurement)
C = clearance volume (determined by volume of oil required to fill the clearance space)

60.2 Engine break-in. The following engine break-in schedule shall be run using the 25:1 fuel/oil test mixture:

Duration	Speed
15 minutes	idle
15 minutes	3000 rpm
15 minutes	1500 rpm
15 minutes	3500 rpm
15 minutes	2000 rpm
15 minutes	4000 rpm
15 minutes	2500 rpm
15 minutes	full throttle, record rpm

Full throttle speed on test must be full rated speed ±200 rpm. The test wheel shall be modified to provide this speed if it is not within manufacturer's specifications during the last 15 minutes of break-in. After break-in, a new set of spark plugs shall be installed and spark timing checked and reset if necessary to manufacturer's specifications at full rated speed.

60.3 Carburetor adjustment. The carburetor(s) shall be adjusted in a one-hour period prior to the test start-up according to the following procedure:

60.3.1 Warm-up engine at idle for 5 minutes.

60.3.2 Set idle speed at 700 rpm and adjust needle for maximum engine speed maintaining sufficient fuel to avoid backfiring. The range of needle adjustment (rich to lean) should be from 1/2 to 1-1/2 turns. The needle angle or needle size may be changed to attain the above range. If maximum engine speed is obtained over a range of needle setting, set the needle to the mid point of the range.

60.3.3 Re-adjust engine speed to 700 rpm with no change in the needle setting of 60.3.2.

60.3.4 Set engine throttle to full power and adjust test wheel to obtain manufacturer's rated speed. Open high speed needle(s) until rough running is observed. Allow engine to stabilize at this point for one minute then record the needle position, engine speed, spark plug gasket temperature's fuel rate and water-out temperature.

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60.3.5 Close high speed needles 30 degrees and after a one minute stabilization record the data in 60.3.4.

60.3.6 Repeat 60.3.5 sufficiently to define a 5°F. spark plug temperature drop on each side of a maximum temperature and a 100 rpm drop on each side of a maximum engine speed. The needle setting to be used for the test shall be the best compromise between maximum spark plug gasket temperatures and maximum engine speed. Should there be a wide mixture variation such as the richest cylinder cannot be leaned to a high temperature peak the engine shall not be used without correcting the deficiency.

60.4 Test cycle and operating conditions. The basic test cycle is one hour in length, consisting of 5 minutes operation at idle followed by 55 minutes of continuous wide open throttle operation. The following parameters are to be observed, recorded, and maintained within the prescribed limits.

	Idle	Full throttle
Time	Record	Record
Test hours	Record	Record
Engine speed	700 ± 100 RPM	Full Rated Speed ± 200 RPM
Fuel flow	Not recorded	Every-other cycle
Fuel temperature	Not recorded	Recorded during flow measurement
Water-in temperature	Record	Record
Water-out temperature	Record	150 ± 5°F.
Tank temperature	Less than 110°F.	
Room temperature	Record	Record
Spark plug gasket temperature	Record	Record
Wet bulb temperature	During first hour of each seven hour cycle	
Barometric Pressure		

60.4.1 The data specified in 60.4 is to be taken each hour at the end of idle cycle, 5 minutes after return to full throttle, and at the end of full throttle operation, unless otherwise specified.

60.4.2 The engine shall be shut down for one hour minimum after 7 one-hour cycles or after each 7 hours of operation.

60.4.3 The 25:1 fuel-oil mix shall be used during break-in and for the first 7 hours of operation. The 50:1 fuel-oil mix shall be used from test hour 8 thru test hour 98. Following the intermediate inspection at 98 test hours, the 100:1 fuel-oil mix shall be used, until conclusion of the test after 147 hours of operation.

60.4.4 Compression pressure. Measure and record the compression pressure of each cylinder at the following times:

(a) After carburetor adjustment and prior to start-up of test.
(b) Prior to 98 hour intermediate inspection.
(c) At end of test.

60.4.5 Carburetor adjustment. The carburetor adjustment established in 60.3 shall be confirmed every 7 hours of engine operation.

60.4.6 Ignition timing. The proper ignition timing setting shall be confirmed every 7 hours of engine operation.

60.4.7 Record any periods of engine missing or rough running.

60.4.8 Spark plug fouling. Replace spark plugs when fouled. Retain fouled spark plugs and tag indicating the engine hours when removed. The temperature recorder also should be marked when spark plug changes are made. The following symptoms are generally associated with spark plug fouling:

(a) Inability to restart engine after stalling.
(b) Missing or rough running at high or low speeds.
(c) Large drop in spark plug gasket temperature.
(d) Drop in engine speed.

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60.4.9 Pre-ignition. Pre-ignition is generally indicated by a sudden rise in spark plug temperature and drop in engine speed. Record all occurrences of pre-ignition including any resulting engine damage.

60.4.10. Lower unit. The lower unit should be drained of any water and filled with new lube every 7 hours of engine operation. Otherwise, lubrication of the lower unit should be in accordance with the manufacturer's recommendations.

70. ENGINE INSPECTION

70.1 Intermediate inspection. After 98 hours of operation, the test engine shall be removed from the tank for intermediate inspection. After checking the compression pressure in each cylinder remove cylinder head(s), intake port covers, and the exhaust port cover(s).

70.1.1 Estimate the overall piston varnish by rating the areas of the piston skirts visible through the intake and exhaust ports as follows:

70.1.1.1 Intake side. Rotate the crankshaft so that the bottom compression ring is just out of sight above the top of the intake ports. Rate the areas exposed by the ports for varnish, using the CRC varnish rating scale. Each area is to be rated individually. Make a mark on the piston skirt with a pencil, tracing along the bottom of one of the intake ports. Rotate the crankshaft so that the pencil mark made is just out of sight above the top of the intake port. Again rate each skirt area exposed. Again mark the piston as described above with a pencil. Rotate the crankshaft as above to bring the next lower areas into view. The piston should be at or near top dead center at this time. Again rate the exposed areas individually for varnish. Average the ratings obtained to determine the rating for the intake side skirt of the piston.

70.1.1.2 Exhaust side. Rotate the crankshaft so that the bottom compression ring is just out of sight above the top of the exhaust ports. Rate the piston areas exposed by the ports individually for varnish. Since the exhaust ports are oval, and longer than the intake ports, the piston need not be marked. Rotate the crankshaft until the piston is at top dead center, and rate the lower skirt areas exposed by the exhaust ports. Average the ratings obtained to determine the rating for the exhaust side skirt of the piston.

70.1.1.3 The intake side and exhaust side ratings are then averaged to obtain the overall piston varnish estimated rating.

70.1.2 Piston ring sticking. Estimate piston ring sticking by rotating the crankshaft until the rings are visible in the ports, and then depressing each ring with a rounded wood dowel. Ring sticking is to be reported according to the scale in 70.2.1.11. Also evaluate condition of ring face.

70.1.3 Exhaust port restriction. Estimate the percent restriction of each exhaust port. Also, report average restriction of all exhaust ports.

70.1.4 Exhaust passage restriction. Estimate average percent restriction of exhaust passages.

70.1.5 Photographs. Color photographs at least 5 by 7 inches shall be made of the combustion chamber piston crowns, cylinder heads, and exhaust ports and passages.

70.2 Final inspection.

70.2.1 At the conclusion of the test after measuring the compression pressure of each cylinder, disassemble the engine and rate the following:

70.2.1.1 Rate piston varnish using CRC Varnish Scale for both thrust and anti-thrust sides.

70.2.1.2 Rate piston scuffing in percent area scuffed.

70.2.1.3 Rate piston crown deposits as percent area of light deposit (1/64 inch), medium deposit (1/32 inch), and heavy deposit (1/16 inch).

70.2.1.4 Rate ring groove deposits as percent volume filled of soft, medium, or hard carbon using the CRC designation for these categories.

70.2.1.5 Rate cylinder head deposits as specified in 70.2.1.3.

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70.2.1.6 Rate exhaust port and exhaust passage restriction as specified in 70.1.3 and 70.1.4, respectively.

70.2.1.7 Rate condition of each piston ring by estimating percent of face area scuffed or scored.

70.2.1.8 Rate wear of all piston rings by difference in initial and final weight and difference in initial and final butt gap.

70.2.1.9 Rate wear of all cylinders by difference in initial and final bore measurements (see 60.1.2).

70.2.1.10 Rate condition of each cylinder by estimating percent area smooth, polished, scuffed, scratched, grooved, flaked, or chipped.

70.2.1.11 Rate condition of all power head bearings by estimating amount of scratching, pitting, or other abnormal condition. Also rate amount and degree of bearing discoloration.

70.2.1.12 Rate piston ring sticking in accordance with the following definitions and rating scale:

Free ring: A free ring falls because of its weight as a piston held on its side is inverted.
Sluggish ring: A sluggish ring does not fall freely but yeilds to slight pressure.
Pinched ring: A pinched ring is held firmly in the groove but shows no evidence of blowby across its face; such rings have also been referred to as "cold stuck".
Stuck ring: A stuck ring is held firmly in the groove and shows blowby streaks across its face; such rings have also been referred to as "hot stuck".

Rating scale	Description of Ring Condition
10.1	100 percent free
9.5	100 percent sluggish
9.0	0 to 25 percent pinched
8.5	0 to 25 percent pinched with remainder sluggish
8.0	25 to 50 percent pinched
7.5	25 to 50 percent pinched with remainder sluggish
7.0	50 to 75 percent pinched
6.5	50 to 75 percent pinched with remainder sluggish
6.0	75 to 100 percent pinched
5.5	0 to 25 percent stuck
5.0	0 to 25 percent stuck with remainder sluggish or pinched
4.0	25 to 50 percent stuck
3.0	25 to 50 percent sutck with remainder sluggish or pinched
2.0	50 to 75 percent stuck
1.0	50 to 75 percent stuck with remainder sluggish or pinched
0.0	75 to 100 percent stuck

70.2.1.13 Photographs. A color photograph at least 5 by 7 inches shall be made of each of the following engine parts:

(a) Piston (thrush and anti-thrust sides)
(b) Piston rings (only if broken or other abnormal condition is noted)
(c) Cylinder head
(d) Cylinder block showing exhaust ports and passages. (Include similar photograph of a new cylinder block for comparison)
(e) Bearings, main and connecting rod
(f) Crankshaft

70.2.1.14 Engine parts. Retain the engine parts listed in 70.2.1.13 with the exception of the crankshaft.

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80.0 REPORT

80.1 The report of test results shall contain the following engine performance items.

(a) Summary of test
(b) Summary of engine operation including control of test parameters by minimum, maximum, and average.
(c) Determination of compression ratio.
(d) Analysis of test fuel including TEL content, vapor, pressure, octane no., distillation and hydrocarbon analysis.
(e) Engine measurements (see 60.1.2).
(f) All intermediate and final inspection data and summary thereof.
(g) Photographs as designated in 70.1.5 and 70.2.1.13.
(h) Number of spark plugs fouled out and time of fouling.
(i) Number of occurrences of pre-ignition.

80.2 The test report shall also contain the results of tests for determination of conformance with the other requirements contained in Section 3 of this Specification.

80.3 Retain the temperature recorder charts and one copy of the engine test log for presentation with the test report.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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